WORKED SOLUTION

STEP I 1990, Question 1

Pure • Trigonometric Optimisation

STEP I 1990, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s1-q1

The key idea

Every vertex of the blind is at distance r from the centre D, so the radii DX and DY cut the quadrilateral into three isosceles triangles with apex D. Each has area $\frac{1}{2}r^2\sin(\text{apex angle})$, and the three apex angles add up to π. That turns a geometry problem into maximising a sum of sines, and a sum-to-product formula then does the first part without any calculus.

Solution

The area of the blind

Since A, D, C are collinear, the angles at D satisfy

$$\angle ADX + \angle XDY + \angle YDC = \pi, \qquad \text{so} \qquad \angle YDC = \pi - \theta - \alpha.$$

Here $\theta \geq 0$, $\alpha > 0$ and $\theta + \alpha \leq \pi$, so that X and Y lie on the arc in the order A, X, Y, C.

The radii DX and DY split the blind $AXYC$ into triangles ADX, XDY and YDC, each with two sides of length r meeting at D. Using area $= \frac{1}{2}ab\sin C$ for each,

$$\begin{aligned}\text{Area} &= \tfrac{1}{2}r^2\sin\theta + \tfrac{1}{2}r^2\sin\alpha + \tfrac{1}{2}r^2\sin(\pi-\theta-\alpha)\\ &= \tfrac{1}{2}r^2[\sin\alpha + \sin\theta + \sin(\theta+\alpha)],\end{aligned}$$

since $\sin(\pi - u) = \sin u$.

The best θ for fixed α

With α fixed, only $\sin\theta + \sin(\theta+\alpha)$ varies. By the sum-to-product formula $\sin P + \sin Q = 2\sin\frac{P+Q}{2}\cos\frac{P-Q}{2}$,

$$\sin\theta + \sin(\theta+\alpha) = 2\cos\tfrac{\alpha}{2}\,\sin\left(\theta + \tfrac{\alpha}{2}\right).$$

Ask yourself Why does this make the maximum obvious?

The factor $2\cos\frac{\alpha}{2}$ is a fixed positive constant, because $0 < \frac{\alpha}{2} < \frac{\pi}{2}$. So the area is largest exactly when $\sin(\theta + \frac{\alpha}{2})$ is largest. As θ runs over $[0, \pi - \alpha]$, the angle $\theta + \frac{\alpha}{2}$ runs over $[\frac{\alpha}{2}, \pi - \frac{\alpha}{2}]$, which contains $\frac{\pi}{2}$. So the sine reaches its greatest possible value 1, when $\theta + \frac{\alpha}{2} = \frac{\pi}{2}$, that is,

$$\boxed{\theta = \frac{\pi}{2} - \frac{\alpha}{2} = \frac{\pi - \alpha}{2}.}$$

Then $\angle YDC = \pi - \theta - \alpha = \frac{\pi-\alpha}{2} = \theta$ as well: the best blind is symmetric about the vertical radius, which is what intuition suggests. The maximum area for this α is

$$\tfrac{1}{2}r^2\left[\sin\alpha + 2\cos\tfrac{\alpha}{2}\right].$$

Letting α vary

Now we maximise $f(\alpha) = \sin\alpha + 2\cos\frac{\alpha}{2}$ for $0 < \alpha < \pi$. (For each α the largest area is $\frac{1}{2}r^2 f(\alpha)$, so the overall largest area is $\frac{1}{2}r^2$ times the largest value of f.) Differentiating,

$$f'(\alpha) = \cos\alpha - \sin\tfrac{\alpha}{2}.$$

Write $s = \sin\frac{\alpha}{2}$, so $0 < s < 1$, and use $\cos\alpha = 1 - 2\sin^2\frac{\alpha}{2}$:

$$f'(\alpha) = 1 - s - 2s^2 = (1 - 2s)(1 + s).$$

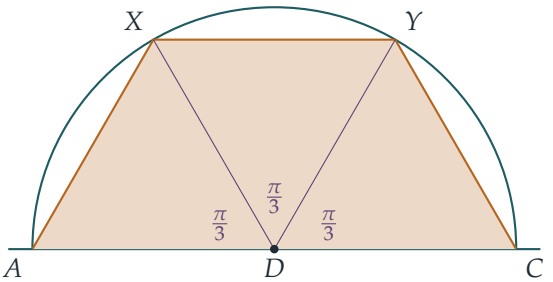
Since $1 + s > 0$, f' has the sign of $1 - 2s$. As s increases with α on $(0, \pi)$, f' is positive for $s < \frac{1}{2}$ and negative for $s > \frac{1}{2}$. So f increases and then decreases, with its greatest value where $\sin\frac{\alpha}{2} = \frac{1}{2}$, that is $\alpha = \frac{\pi}{3}$. There

$$f\left(\tfrac{\pi}{3}\right) = \sin\tfrac{\pi}{3} + 2\cos\tfrac{\pi}{6} = \tfrac{\sqrt{3}}{2} + \sqrt{3} = \tfrac{3\sqrt{3}}{2},$$

so the maximum possible area of the blind is

$$\boxed{\tfrac{1}{2}r^2 \cdot \tfrac{3\sqrt{3}}{2} = \frac{3\sqrt{3}}{4}r^2.}$$

At this point $\alpha = \theta = \angle YDC = \frac{\pi}{3}$, so the three triangles are equilateral and the blind is half of a regular hexagon inscribed in the full circle.



Checks

- Limiting cases: as $\alpha \to 0$ the blind becomes the triangle AXC with X at the top, area $\frac{1}{2} \cdot 2r \cdot r = r^2$, and indeed $\frac{1}{2}r^2 f(0) = r^2$. As $\alpha \to \pi$ the blind collapses onto the diameter, and $f(\pi) = 0$. The maximum $\frac{3\sqrt{3}}{4}r^2 \approx 1.299r^2$ beats both, and is less than the semicircle's area $\frac{\pi}{2}r^2 \approx 1.571r^2$, as it must be.
- Numerically: we checked the three-triangle formula against the shoelace formula for random positions of X and Y, located the best θ by a fine grid search for several values of α, and found that a joint search over θ and α with $r = 1$ gives largest area $1.299038\ldots = \frac{3\sqrt{3}}{4}$.

Common slips

- Forgetting the third triangle YDC, or giving its angle as $\pi - \alpha$ rather than $\pi - \theta - \alpha$.
- Using the shortcut $\theta = \alpha = \frac{\pi}{3}$ in the second part without justifying it. The calculus (or a symmetry argument) is needed to show that the best blind really is half of a regular hexagon.

Beyond STEP

The answer is a special case of a classical fact: among all n-gons inscribed in a circle, the regular one has the greatest area. One neat proof uses the concavity of sin on $[0, \pi]$ and Jensen's inequality: if the central angles $\phi_1, \ldots, \phi_n$ add to 2π, then $\sum \sin \phi_i \leq n \sin \frac{2\pi}{n}$. Our blind is half of the case $n = 6$. Convexity and inequalities of this kind are a running theme in first-year analysis; Abbott's *Understanding Analysis* is a good place to meet them properly, and Siklos's *Advanced Problems in Mathematics* has more STEP problems in the same spirit.

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